

Jiahui Du

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RESEARCH INTERESTS

Human-Computer Interaction (HCI), Accessible Computing, Multimodal Interaction, VR/XR, Computer Vision, Human-Centered AI

EDUCATION

University of Illinois Urbana-Champaign
Master of Computer Science

Urbana, IL
Aug 2026 – Expected May 2028

Xi'an Jiaotong-Liverpool University

B.S. in Information and Computing Science, *First-Class Honours*; GPA: **3.86/4.0**

Suzhou, China
Sep 2022 – Jul 2026

TECHNICAL SKILLS

Programming: Python, Java, C#, SQL, JavaScript; **AI/ML:** PyTorch, scikit-learn, CNNs, transfer learning, clustering
HCI/XR: Unity, VR development, prototyping, usability testing, heuristic evaluation, user-centered design
Computer Vision/Graphics: object detection/tracking, 2D pose estimation, camera calibration, OpenGL, 3D transformations
Software/Data: Vue 3, Spring Boot, REST APIs, MySQL, web scraping, data cleaning, visualization

RESEARCH & SELECTED PROJECT EXPERIENCE

Generating New Perspectives for Watching Sports in Virtual Reality

Undergraduate Dissertation; VR, Computer Vision, Unity, 3D Reconstruction

- Developed a proof-of-concept pipeline that transforms ordinary **single-camera table-tennis footage** into an explorable **3D replay environment** for immersive viewing in Unity/VR.
- Used **table detection and camera calibration** to establish a table-centered coordinate system, and combined **ball detection, segmentation, and tracking with 2D human pose estimation** to recover ball trajectories and player skeletal motion.
- Converted reconstructed ball and player data into a common spatial coordinate system and imported the results into Unity to support replay from viewpoints beyond the original fixed camera perspective.
- Evaluated the prototype in terms of **functional completeness, reconstruction stability, and user experience**, and analyzed limitations including player occlusion, motion blur, and camera-calibration error.

AI-Driven Multi-Perspective VR Storytelling System

SURF Research; VR, Deep Learning, Viewpoint Reconstruction, NLP

- Conducted a Summer Undergraduate Research Fellowship project on an interactive VR storytelling system that enables users to experience the same narrative from multiple characters and viewpoints rather than from a fixed perspective.
- Participated across the research lifecycle from **literature review and system conceptualization to implementation**; later served as a **technical/team lead** contributing to system design and core technical integration.
- Explored deep-learning viewpoint reconstruction, style-transfer/NLP techniques for character transitions, speech generation, and AI-based inference for missing narrative events; selected from **585 submissions** and received **Excellent Poster, School Winner, and Student-Nominated Winner**.

Wander Genius

AI-Powered Personalized Travel Planning Application

HCI, UX Research, Prototyping, User Testing

- Worked in a five-person HCI team through a complete **user-centered design lifecycle**: requirements discovery, data collection, alternative designs, prototyping, usability evaluation, and iterative refinement.
- Analyzed **239 valid questionnaire responses** across age groups to identify information-filtering and route-planning pain points, translating user needs into features including voice interaction and accessibility-oriented functions.
- Served as **prototype-design lead**, driving four low-/mid-/high-fidelity iterations and refining interaction design using interviews, user feedback, Nielsen's usability heuristics, Shneiderman's design rules, and preference testing.

Interactive 2D/3D Computer Graphics Systems

OpenGL, 3D Interaction, Hierarchical Modelling, Camera Systems

- Built interactive 2D and 3D graphics environments with parametric geometry, transformations, animation, lighting, collision detection, and physics-based motion.
- Implemented articulated character animation and **first-person, third-person, and overhead camera systems** with perspective/orthographic projection, strengthening experience with spatial interaction and 3D scene design.

RELEVANT WORK EXPERIENCE

CRRC Qishuyan Institute Co., Ltd. Software Development Intern	Changzhou, China Jul 2025 – Sep 2025
- Contributed to computer-vision algorithms for detecting safety helmets and reflective vests in workshop environments, refining the recognition pipeline through algorithm adjustments. - Designed and implemented a visual frontend for detection results and alerts, and supported technical documentation covering algorithm logic, interface design, and system integration.	
Jiangsu Xinyouchang Information Technology Co., Ltd. Implementation Consultant	Changzhou, China Jun 2024 – Sep 2024
Developed and optimized SQL queries and configured ERP systems for client deployments, translating operational requirements into working software configurations.	

ADDITIONAL TECHNICAL PROJECTS

CIFAR-10 Image Classification with CNN and Transfer Learning	<i>Python, PyTorch, ResNet-50, CUDA</i>
Built and evaluated an end-to-end image-classification pipeline; improved test accuracy from 56.9% to 79.7% and Macro F1 from approximately 0.57 to 0.80 using an ImageNet-pretrained ResNet-50 and training refinements.	
Machine Learning Classification and Clustering Analysis	<i>Python, scikit-learn, PCA/LDA/t-SNE, Ensemble Learning</i>
Implemented supervised and unsupervised pipelines with dimensionality reduction, ensemble classification, and GMM/K-means/hierarchical/DBSCAN clustering; evaluated models using F1-score and silhouette analysis.	
AIGC Recruitment Data Analysis and Business Application Study	<i>Python, Web Scraping, Data Cleaning, Visualization</i>
Built the project’s end-to-end data workflow, scraping and integrating recruitment and company data into a balanced dataset of 234 AIGC companies and 234 traditional IT companies; analyzed employment, company, and text-based business characteristics to support evidence-based recommendations.	

SELECTED AWARDS & ACTIVITIES

- Competitive Table Tennis:** Singles 5th / Team 5th, 2024 Suzhou City 16th Games; Women’s Singles Champion and Team Overall Points Champion, 2023 XJTU College Cup; Women’s Team 5th, 2021 Jiangsu Provincial Table Tennis Championships.
- Research Recognition:** Excellent Poster, School Winner, and Student-Nominated Winner for the SURF multi-perspective VR storytelling project.

LANGUAGES

Chinese (Mandarin) – Native; English – Advanced proficiency